Web Development Foundations

Week 6 – Debugging: "Things Break - And I Can Fix Them"

Day 2: "Debugging Multiple Issues"

1

Reconnect

Monday:

- reproduced one problem
- read one console clue
- compared HTML and JavaScript
- fixed one selector
- verified the result

Today:

- we keep that process when more than one issue appears.

2

One Bug To Issue List

3

The Working Problem: More Than One Thing Can Be Wrong

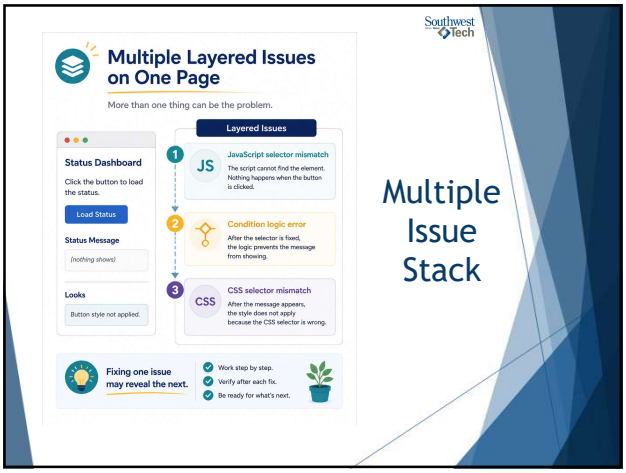
A page can have multiple issues:

- JavaScript selector mismatch
- condition logic error
- CSS selector mismatch
- confusing user feedback

Fixing one issue may reveal the next.

That is normal.

4



Multiple Issue Stack

5

What Counts as Success Today

- you choose one likely cause first
- you make one focused change
- you retest before moving on
- you document what happened
- at least two fixes are verified

6

Prioritize The First Cause

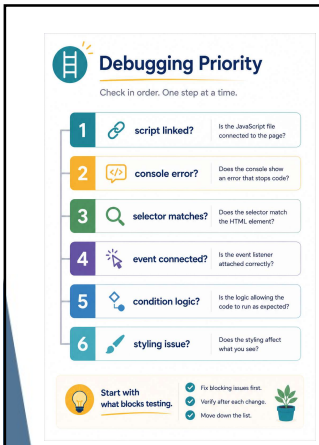
Start with the issue that blocks testing.

Good first checks:

- script file linked?
- console error present?
- selector matches HTML?
- event listener connected?

Cosmetic issues can wait until behavior works.

7



Priority Ladder

8

What We Will Use Today

- issue list
- priority
- console error
- `console.log()`
- selector check
- condition check
- CSS mismatch check
- verification note

9

Debugging Toolbox

10

Do Not Fix Everything At Once

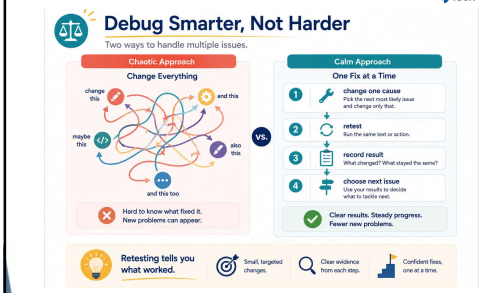
Avoid This

- change selector
- change condition
- change CSS
- refresh once
- hope it works

Use This

- change one likely cause
- retest
- record result
- then choose the next issue

11



One Fix At A Time

12

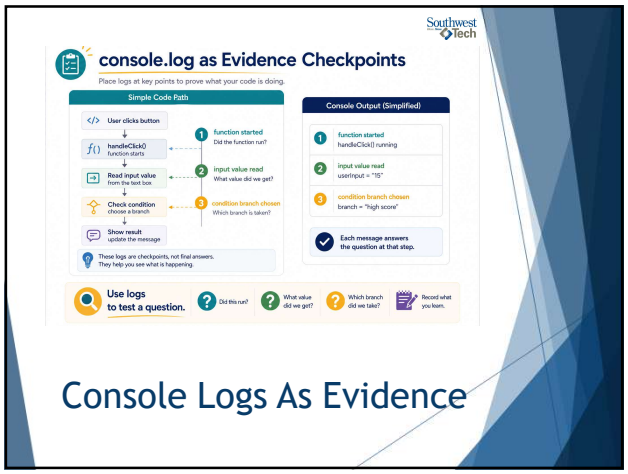
Console Logs As Evidence

`console.log()` can answer:

- did this function run?
- what value did the input contain?
- which branch did the condition choose?
- did the code reach this line?

Use logs to test a question.
Then remove or clean them later.

13



Console Logs As Evidence

14

Proper `console.log()` Format

Use logs that explain what you are checking.

Less useful:

- `console.log(taskText);`

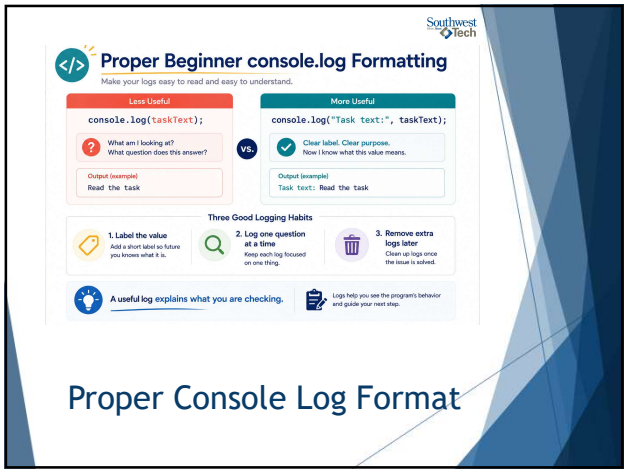
More useful:

- `console.log("Task text:", taskText);`

Best habit:

- - label the value
- - log one question at a time
- - remove extra logs after debugging (or comment them out)

15



Proper Console Log Format

16

Demo

"Multi-Issue Debugging"

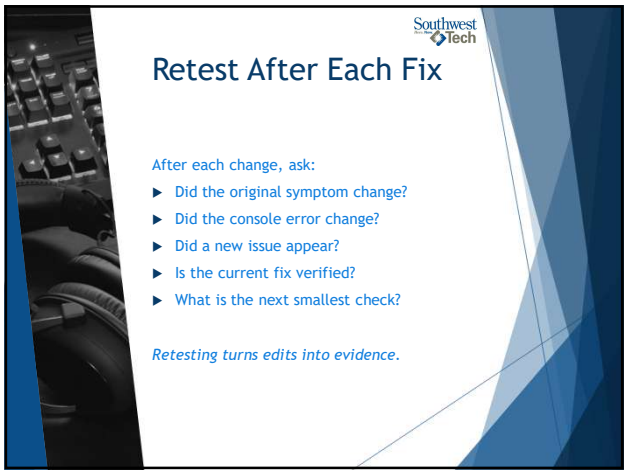
17

What to Watch For

- first visible symptom
- first console clue
- selector mismatch
- retest after selector fix
- condition error
- CSS issue handled after behavior works

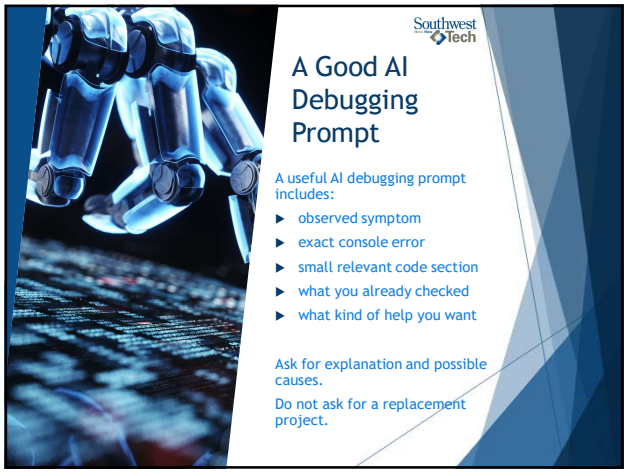
18

19

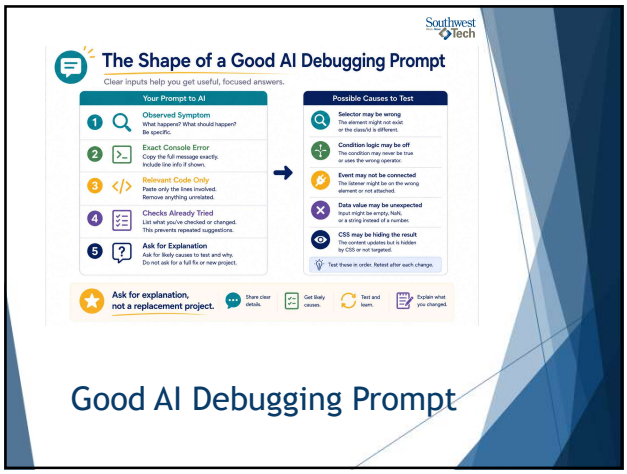


20

21

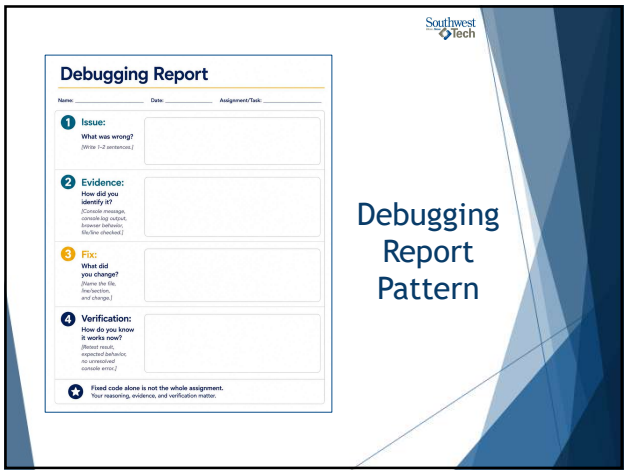


22



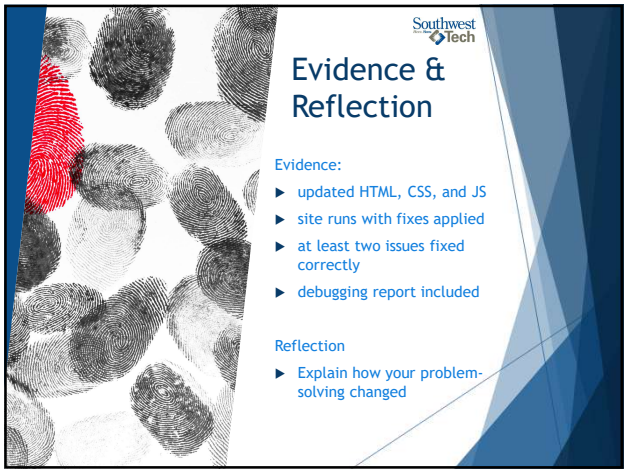
23

24

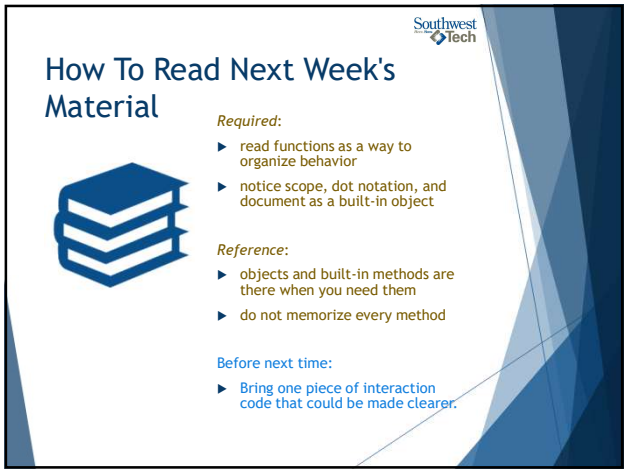


25

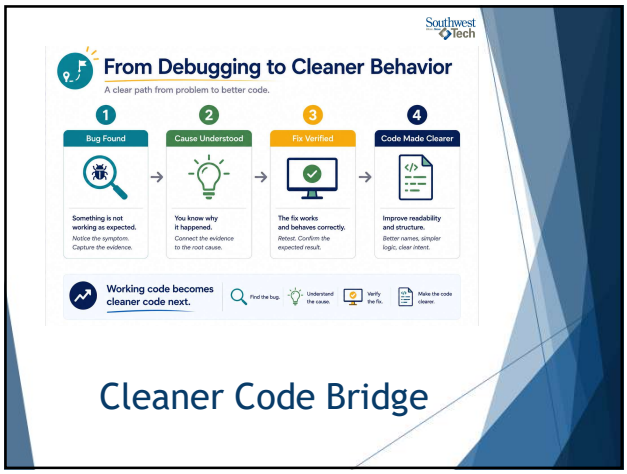
26



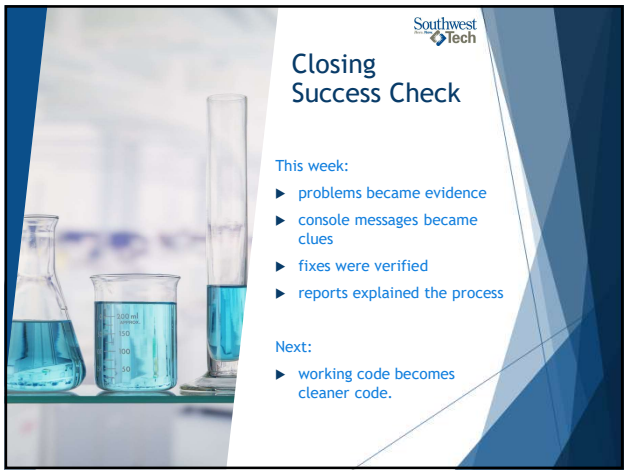
27



28



29



30

31